

WASP-43b

R-1983

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_201230 · created 2026-08-07T01:03:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459213.9048 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.045
Transit depth (Rp/Rs)^2	3.68 +/- 1.74 [%]
Orbital Inclination (inc)	79.02 +/- 2.3
Airmass coefficient 1 (a1)	1.011 +/- 0.022
Airmass coefficient 2 (a2)	-0.006 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.14 %
Best Comparison Star	#2 - [258, 62]
Optimal Method	PSF photometry
Transit Duration (day)	0.037 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.192 ± 0.045 vs 0.1594 (deviation 0.72σ)
Duration fit vs published	0.037 d vs 0.0512 d (deviation 0.89σ)
Mid-time O–C	-2.1 min
Depth SNR	4.3
Residual scatter	2.14 %

Light curve

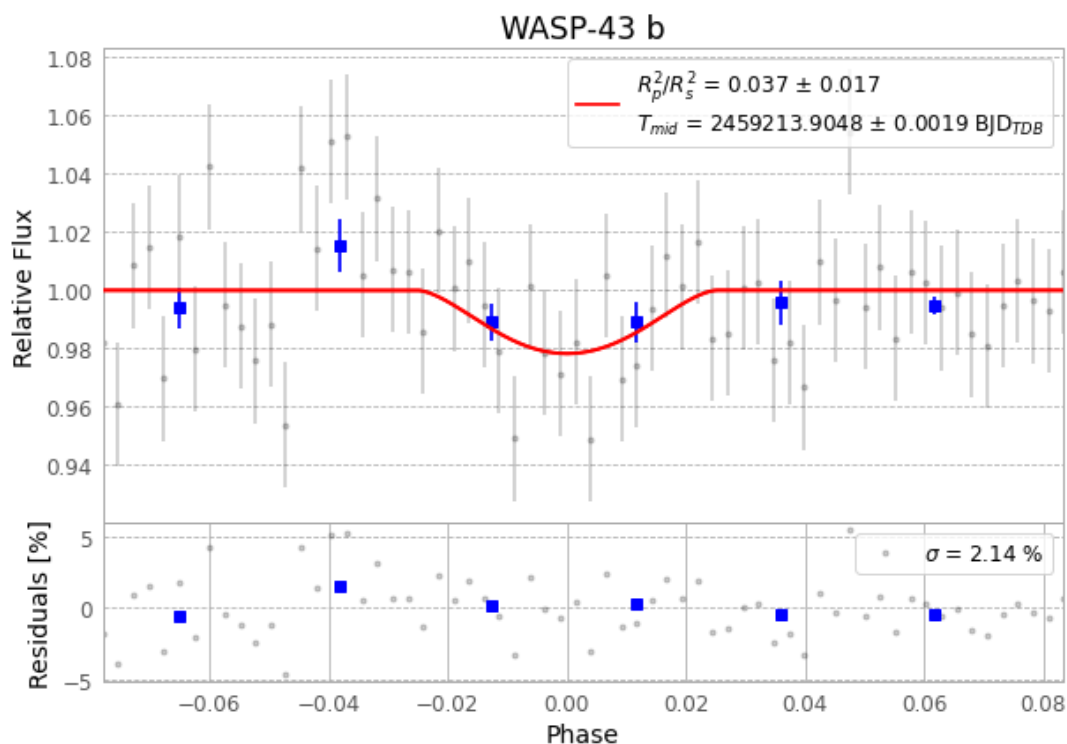


Figure 1 — FinalLightCurve_WASP-43 b_2020-12-30.png (SHA-256 e29bdc6825c71bb2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 245], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6e0cd28467124207...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), WASP-43201230080612.FITS → WASP-43201230111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-201230.log (b2c69ee7c4b0...), FinalLightCurve_WASP-43 b_2020-12-30.png (e29bdc6825c7...), FinalLightCurve_WASP-43 b_2020-12-30.csv (cf795739852e...)

Compute resources

Wall time 97.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 01:03Z.